Digital Signal Processing

1. a) Compare and contrast FIR and IIR filters. [4]

Solution:
Explanation of structural differences, the role of feedback;
Explanations in terms of poles and zeros;
Stability considerations;
Computational complexity.

This part was done successfully by most students.

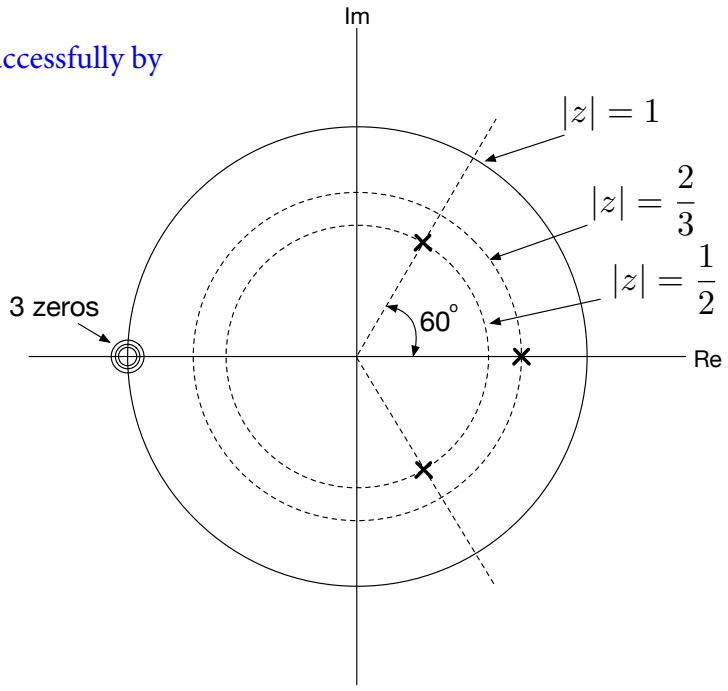


Figure 1.1 Pole-zero diagram

b) The pole-zero plot of a digitial filter is given in Fig. 1.1. The gain of this filter at d.c. is 1.

i) Find the system function of this filter in the form

$$H(z) = A\left[\frac{(1+a_1z^{-1})(1+b_1z^{-1}+b_2z^{-2})}{(1+c_1z^{-1})(1+d_1z^{-1}+d_2z^{-2})}\right]$$

giving the values of the constant coefficients A, a_1, b_1, b_2, c_1, d_1, d_2.

[8]

This part cause some mathematical difficulties for some students. Only a few simplified the fraction.

Solution:

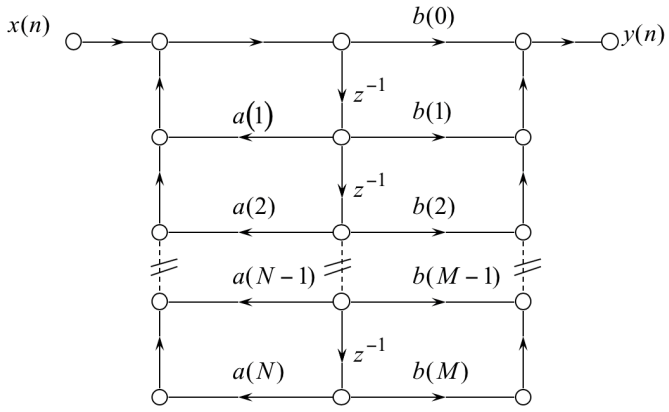
$$H(z) = A\left[\frac{(z+1)^3}{(z-\frac{2}{3})(z-(\frac{1}{4}+j\frac{\sqrt{3}}{4})(z-(\frac{1}{4}-j\frac{\sqrt{3}}{4}))}\right]$$
$$= A\left[\frac{(1+z^{-1})(1+2z^{-1}+z^{-2})}{(1-\frac{2}{3}z^{-1})(1-\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2})}\right]$$

Therefore $a_1 = 1$, $b_1 = 2$, $b_2 = 1$, $c_1 = -\frac{2}{3}$, $d_1 = -\frac{1}{2}$, $d_2 = \frac{1}{4}$. For unit gain at d.c. we required $A = 32$.

c) Draw a signal flow graph for the filter of part b) in Direct Form II. [4]

Well done.

Solution:



d) Discuss the impact on the filter's pole-zero plot that would result if the coefficient d_2 was found to be erroneously increased by 10%. [4]

Solution: The impact of an error in the denominator coefficients may be more significant than an error in the numerator coefficients due to error feedback arising in the recursive structure.
The increase from $d_2 = 0.25$ to $d_2' = 0.275$ moves the roots in polynomial d in modulus from 0.5 to 0.52 and in angle from ± 1.05 to ± 1.07.

As expected, only students with deeper insights were able to answer this part adequately. Correct quantitative answers were rare.

2. a) Consider the system function given by

$$H(z) = \sum_{k=0}^{\infty} b_k z^{-k}.$$

Describe the characteristics of the filter represented by this system function and state the requirements that must be satisfied for the filter to be causal and stable. [4]

Solution:
This is a nonrecursive (MA) filter. It might be described as 'FIR' but note the semi-infinite support.
In general a system $H(z)$ is stable if its ROC includes the unit circle in the z-plane. The ROC for this filter structure excludes $z = 0$ and therefore includes the unit circle. Accordingly, nonrecursive systems such as this cannot be unstable. Equally $H(z)$ is stable if its inverse z-transform $h(n)$ satisfies

$$\sum_{-\infty}^{\infty} |h(n)| < \infty.$$

A right-sided impulse response indicates causality.

Well done.

b) Explain the key properties of a linear phase FIR filter. Include discussion in terms of the filter coefficients and in terms of the z-domain. Include relevant sketches to illustrate your explanation. [6]

Solution:
The filter coefficients satisfy $h(n) = h(N-1-n)$. The zeros in the z-domain occur in mirror image pairs such that if a zero exists at z_0 then another zero also exists at $\frac{1}{z_0^*}$. Sketches are required for the filter coefficient symmetry and an example z-plot plot.

Only a few students demonstrated that they didn't know anything about linear phase FIR filters.

c) Show that the frequency response of a linear phase FIR filter with N coefficients, N even, can be written as

$$H(e^{j\omega}) = e^{j\lambda} \sum_{n=0}^{M-1} 2h(n)\cos(\gamma)$$

and give expressions for M, λ and γ. [5]

Solution:
This follows from

$$H(z) = \sum_{n=0}^{N/2-1} h(n)\left[z^{-n} + z^{-(N-1-n)}\right]$$

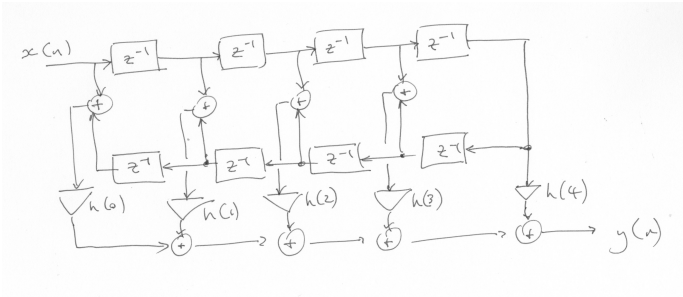
so that $M = N/2$, $\lambda = -\omega[(N-1)/2]$, $\gamma = \omega(n - \frac{N-1}{2})$.

Well done by the majority. As usual, several students tried to 'reverse engineer' the solution and usually were not convinving.

d) Draw and label a signal flow diagram of the efficient implementation of a linear phase FIR filter for $N = 9$. [5]

Those that did this did it well. I was surprised that many students omitted this part.

Solution:



3. Consider an LTI system operating with sampling frequency 10 kHz and system function

$$H(z) = z^{-1}.$$

a) Draw labelled sketch graphs of the magnitude and phase of the frequency response of this system. [2]

Well done mostly.

Solution: The sketch graphs should clearly show a gain of 1 or 0 dB across frequency with a corresponding phase of $-\omega$, linear with frequency. Axes must be correctly labelled to obtain the marks.

b) Write down the general expression for group delay and give the units of group delay. Determine the group delay of $H(z)$. [3]

Well done by the majority.

Solution:
Group delay $= -\frac{d\phi}{d\omega}$.
Units of time (s). Samples would also be acceptable.
Group delay of $H(z)$ is 0.1 ms.

c) Consider the vector

$$\mathbf{x} = [x(0)\ x(1)\ \cdots\ x(N-1)]^T$$

the elements of which are samples of the input signal and ^{T} indicates the transpose operation. Further consider the vector

$$\mathbf{y} = [y(0)\ y(1)\ \cdots\ y(N-1)]^T$$

the elements of which are samples of the output signal.

i) Given that the processing due to $H(z)$ is formulated as a matrix operation $\mathbf{y} = \mathbf{H}\mathbf{x}$, write out the elements of $\mathbf{H}$. Use the notation $\cdots$ to account for an undefined value for N. [4]

Parts (c) and (d) were challenging as the method/ approach is not covered specifically in the lectures and requires a good understanding of the processing as well as some degree of competence with matrix algebra. However, a lot of students did really well on this. Perhaps this is an example where doing the matlab coursework would have been of assistance.

Solution:

$$\mathbf{H} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \ddots & & & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

ii) Now write out the elements of $\mathbf{H}_c$ representing the processing due to $H_c(z)$ in matrix form where $H_c(z) = H(z)$ except that any shifts are performed circularly with modulo N. [3]

Solution:

$$\mathbf{H_c} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \ddots & & & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

iii) Given that the DFT is formulated as a matrix operation $\mathbf{y} = \mathbf{F}\mathbf{x}$, write out the elements of $\mathbf{F}$. [3]

Solution:

$$\mathbf{F} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & e^{-j2\pi/N} & e^{-j2\cdot 2\pi/N} & \cdots & e^{-j(N-1)\cdot 2\pi/N} \\ 1 & e^{-j2\cdot 2\pi/N} & e^{-j4\cdot 2\pi/N} & \cdots & e^{-j2(N-1)\cdot 2\pi/N} \\ \vdots & \ddots & & & \vdots \\ 1 & e^{-j(N-1)\cdot 2\pi/N} & e^{-j2(N-1)\cdot 2\pi/N} & \cdots & e^{-j(N-1)(N-1)\cdot 2\pi/N} \end{bmatrix}$$

d) Consider writing the processing due to $H_c(z)$ in matrix form so that

$$\mathbf{H}_c = \mathbf{F}^{-1}\mathbf{A}\mathbf{F}.$$

Find the elements of the matrix $\mathbf{A}$. Comment on the significance of this formulation and state any characteristic properties of the rows, columns and diagonals of $\mathbf{F}$ and $\mathbf{A}$ in this context. Discuss briefly the role of normalization in the DFT. [5]

Solution:

$$\mathbf{A} = N \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & e^{-j2\pi/N} & 0 & \cdots & 0 \\ 0 & 0 & e^{-j2.2\pi/N} & \cdots & 0 \\ \vdots & \ddots & & & \vdots \\ 0 & 0 & 0 & \cdots & e^{-j(N-1).2\pi/N} \end{bmatrix}$$

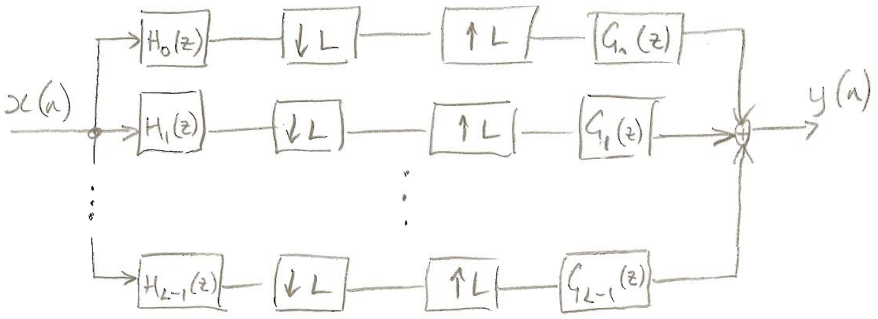
This formulation shows that the DFT diagonalizes the unit circular delay operator so that the columns of $\mathbf{F}$ are eigenvectors and the diagonal elements of $\mathbf{A}$ are the corresponding eigenvalues. The normalization embeds the term $1/N$ usually in the inverse transform to achieve the unitary property in $\mathbf{F}$.

4. A particular multirate digital signal processing system is formed from an analysis filter bank followed by a synthesis filter bank. The analysis and synthesis filter banks both contain L filters indexed $k = 0, 1, \ldots, L-1$ and are denoted respectively $H_k(z)$ and $G_k(z)$.

a) Draw a block diagram of this multirate system and label the input signal $x(n)$, the output signal $y(n)$ and the filters $H_k(z)$ and $G_k(z)$ for arbitrary L. [4]

Well done by most.

Solution:



b) Discuss the advantages and disadvantages of polyphase implementations of filter banks. [2]

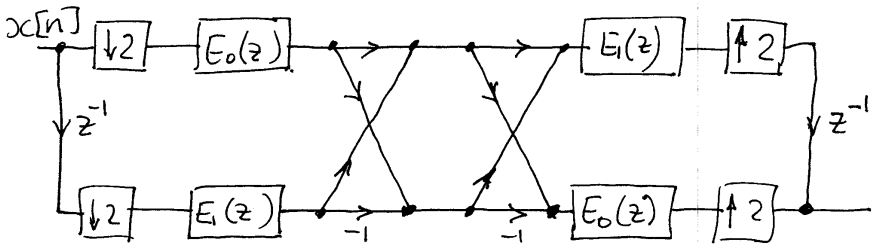
Well done by most.

Solution:
Polyphase filter banks benefit from efficiency in that all filter computations can be performed at the lowest sampling rate in the system.

c) Draw a fully labelled block diagram of the multirate system of a) with $L = 2$ implemented in Type 1 polyphase form, where the polyphase filters are denoted $E_{k\ell}(z)$ for phase ℓ. [3]

Solution:
Each filter in part a) is replaced with it's polyphase representation. 2 marks if the direct substitution is made. 3 marks if an efficient implementation is provided such as

The need to replace *each* filter by its polyphase implementation should have been clearly shown in solutions. The special case as shown here for L=2 leads to this particularly efficient realization.



d) Find the relationship between the coefficients of the filter $E_{k\ell}(z)$ and the coefficients of the filter $H_k(z)$. [3]

Some students omitted this part unfortunately but those that attemted it did well generally.

Solution:
The coefficients of $E_{k\ell}(z)$ are a subset of the coefficients of $H(z)$ obtained by subsampling such that for the kth filter

$$e_{k,\ell}(n) = h_k(nM + \ell)$$

where M denotes the number of phases.

e) Show that the expression for $H_k(z)$ can be written in matrix form as

$$\mathbf{h}(z) = \mathbf{E}(z^L)\mathbf{e}(z)$$

and give the definitions of $\mathbf{h}(z)$, $\mathbf{E}(z)$ and $\mathbf{e}(z)$, where $\mathbf{E}(z)$ is called the polyphase component matrix of the analysis filter bank. [4]

Solution:

$$\mathbf{h}(z) = \mathbf{E}(z^L)\mathbf{e}(z)$$

Well done in many cases.

$$\mathbf{h}(z) = [H_0(z)\ H_1(z)\ \ldots\ H_{L-1}(z)]^T$$
$$\mathbf{e}(z) = [1\ z^{-1}\ \ldots\ z^{-(L-1)}]^T$$
$$\mathbf{E}(z) = \begin{bmatrix} E_{00}(z) & E_{01}(z) & \ldots & E_{0,L-1}(z) \\ E_{10}(z) & E_{11}(z) & \ldots & E_{1,L-1}(z) \\ \vdots & \vdots & \ddots & \vdots \\ E_{L-1,0}(z) & E_{L-1,1}(z) & \ldots & E_{L-1,L-1}(z) \end{bmatrix}$$

f) Consider the matrix $\mathbf{R}(z)$ which is the polyphase component matrix of the synthesis filter bank. Define what is meant by *perfect reconstruction* in this context and deduce the condition on $\mathbf{R}(z)$ for perfect reconstruction. [4]

Solution:

Perfect reconstruction:

I was disappointed that this part did not score well in general. Not sure why as it is a straightforward test of knowledge.

- Alias cancellation
- No amplitude distortion except for a multiplicative arbitrary gain.
- No phase distortion except for an arbitrary delay.

$$\hat{X}(z) = cz^{-n_0}X(z)$$

for scalar constants c, n_0. In matrix form

$$\mathbf{R}(z)\mathbf{E}(z) = \mathbf{I}.$$